write a shell program to implement while loop, print first 10 numbers

```
#!/bin/bash
i=1
while [ $i -le 10 ]
do
echo " $i "
(( i++ ))
Done
```

https://youtu.be/SFIXnmIcmOI?si=_KMzkCHEcaqcmUHH

write a shell program to implement if statement

```
#!/bin/bash
n=10
if [ $n ==  10 ]
then
echo "n is equal to 10"
fi
if [ $n != 10 ]
then
echo "n is not equal to 10"
fi
```

https://youtu.be/OO2Avn1g5Tw?si=ssarH1meqypYLqpX

write a shell program to find the factorial of a number

```
#!/bin/bash

echo "enter a number"

read num


fact=1

for((i=2; i<=num; i++))

{

fact = $((fact*i))

}

echo $fact
```

Write a shell program to check the given year is leap or not

```
#!/bin/bash


# This script checks if the given year is a leap year or not


echo "Enter a year: "

read year


# Check if the year is divisible by 4
```

```
if [ $((year % 4)) == 0 ]; then
  # Check if the year is not divisible by 100
  if [ $((year % 100)) != 0 ];
then
    echo "$year is a leap year"
  # If the year is divisible by 100, check if it is divisible by 400
  else
    if [ $((year % 400)) == 0 ]; then
      echo "$year is a leap year"
    else
      echo "$year is not a leap year"
   fi
  fi
else
  echo "$year is not a leap year"
fi
```

Note: The expr command reads the Expression parameter, evaluates it, and writes the result to standard output.

write a shell program to print table of any number

```
#!/bin/bash
echo "Enter NO.:"

read no

echo "Enter range:"

read r

i=1



while test $i -le $r          #starts with 1 to given range

do

   temp=`expr $no \* $i`

   echo "$no * $i = $temp"

   i=`expr $i + 1`

done
```

Write a shell program for swapping of two nos

```
firstNumber=12

secondNumber=24

echo "Before Swapping"

echo "First number: $firstNumber"

echo "Second number: $secondNumber"

temp=$firstNumber

firstNumber=$secondNumber

secondNumber=$temp

echo "After Swapping"

echo "First number: $firstNumber"

echo "Second number: $secondNumber"
```

Output of the above program

```
Before Swapping
First number: 12
Second number: 24
After Swapping
First number: 24
Second number: 12
```